**Physics 3181 — Spring 2025**

# *Computational Physics*

Lectures: Mon/Wed 2:20-3:35 pm, Corcoran Hall, Room 404A
Instructor: Prof. Oleg Kargaltsev
Office: Corcoran Hall 407; Office hours: Tuesday 3-4:30 pm, or by appointment
Email: kargaltsev@gwu.edu

Prerequisites: MATH 2233 and PHYS 3161, programming experience (Python is mostly sufficient)

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**Short description:** The purpose of this course is to *discuss and practice* basic numerical methods useful for solving Physics, Astrophysics, and Applied Mathematics problems. The course will cover methods that are frequently encountered in these fields. These methods are well established and have a solid theoretical foundation. However, the focus will be on *a practical applications of these methods to solve concrete problems*.

**Details:** The course is project-driven and, typically, a new topic will be discussed every week. To help you understand the material and to provide an assessment of your progress, you will have to solve a homework each week. Often, the homework will be motivated by physics problems. To solve them you will rely on the numerical methods discussed in class. Required physics background will be also briefly reviewed in class.

The primary programming languages for this course are Python *and Wolfram Mathematica*. Some examples may be also given in C. We will use Python (or, possibly, C in some cases) to practice low-level programming and solve some basic computational problems. Mathematica may be used to solve higher-level problems, to assist with debugging the codes and verifying Python (C) code results, to further analyze and visualize the results of computations, to create plots and animations, etc. For each homework set, you will be asked to submit a *carefully documented* code and, in some cases, write a report (in MS Word or Latex; this will allow you to typeset equations, include graphs to support your presentation, as well as to create a bibliography). An effective workflow depends on your ability to use these tools. Although some introduction will be provided, this course is not meant to teach you any programming language or Latex. You will need to fill in any gaps in your programming skills yourself (feel free to use online resources or ask your classmates for help). Students must bring their own laptops to class and configure them appropriately to be able to run Python and Mathematica. The use of Unix (the one in MacOS or some other Linux flavor) is preferred (your instructor will use laptop with MacOS).

I currently prefer using Python over C because Python use has become much more widespread in physics and astronomy although in certain areas of physics C is still the language of choice.

If you are not familiar with Python, I can recommend a brief introduction that can be found here: https://blogs.umass.edu/candela/computational-physics-in-python/?utm_source=pocket_reader . Please review and try running some simple Python code on your laptop *before our course begins*. Please install Python 3.10 or newer and JupyterLab (unless you prefer other Python IDE like Spyder). If you are not very good at installing things it is probably best to install via Anaconda (see https://runestone.academy/ns/books/published/httlads/PythonReview/anaconda.html). Please also install

Wolfram Mathematica (available for free for GW students) before you come to class (see https://it.gwu.edu/wolfram-mathematica).

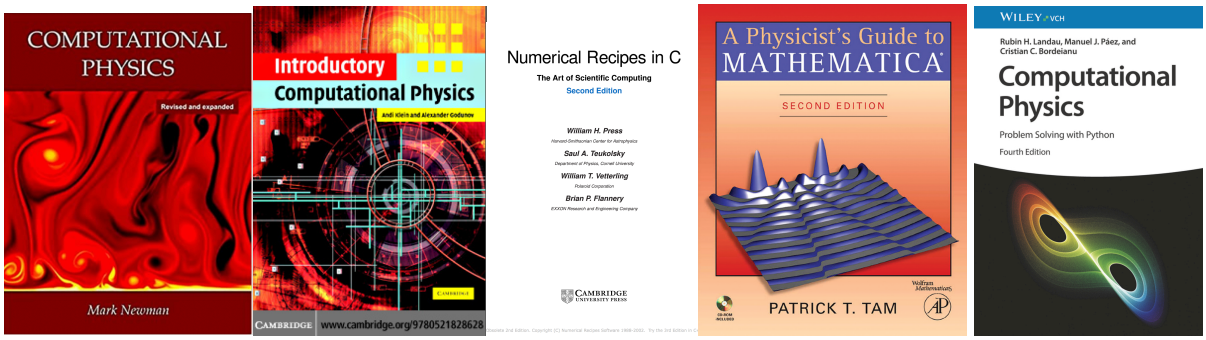
**Learning Outcomes:** Upon completion of this course, you will be able to write programs in Python and Mathematica and apply programming skills in conjunction with the laws of physics to solve a wide range of physics problems.

Students will acquire the following *specific skills* that will be verified via specially designed, graded homework assignments:

- Numerical Integration
- Solving differential equations
- Optimization
- Hypothesis testing and fitting
- Chaos/N-body dynamics
- Machine Learning
- Monte Carlo methods
- Root finding
- Fourier transform

**Textbook:** No textbook is required. The lecture slides and notes will be provided by the instructor as needed. They will be accessible in GW Blackboard.

If you still want to have a textbook, I can recommend:




**Mark Newman "Computational Physics"** (ISBN-13: 978-1480145511)

Klein, Cambridge University Press 2006, "Introductory Computational Physics"

Tam, Cambridge University Press 2006 "A Physicist's Guide to Mathematica"

Numerical Recipes in C available free at http://www.nrbook.com/a/bookcpdf.html

*If you want to purchase only a single book (not required)*, I recommend Python-based Mark Newman's "Computational Physics" (ISBN-13: 978-1480145511) or Rubin Landau et al. "Computational Physics" (ISBN-13: 978-3527414253). The former is simper and smaller, the latter is more comprehensive and includes more complex problems.

**Assignments:** Coding homework projects will be posted regularly at the class web site on Blackboard http://blackboard.gwu.edu. Homework must be uploaded to Blackboard. In general, it

is expected to include: (1) the Word/Latex report with the statement of the problem and algorithm/method description, (2) the actual code that run. *The results of calculations in numerical and/or graphical format and interpretation* of the results by student must be present either in the report or in the code (the latter possible if you are using, e.g., Jupyter notebook). Students must visit the Blackboard site for this course regularly. Term project is a more your own investigation of a more complex physics problem that is either given to you by your instructor or comes from your independent research (but needs to be approved by the instructor as suitable for this course).

**Expected Time Investment:** Over 14 weeks students will spend 3 hours per week in classroom activities (lectures, quizzes, problem-solving and programming projects). Minimum amount of independent, out-of-class, learning expected per week is 8 hrs.

**Outline of the course for Spring 2025**

| Week | Dates | Topic |
|---|---|---|
| **1** | Jan 13, 15 | Introduction and setup. Basics of programming (in Python). Basics of Mathematica. Graphing in Mathematica. Data visualization. Intro to statistics. |
| **2** | Jan 20 | Your first computer simulation: Predator and prey problem. |
| **3** | Jan 22, Jan 27 | Roots of nonlinear equations. Interpolation, cross-correlation, numerical derivative. Sums. Multiple collision accelerator. Resistor networks. |
| **4** | Jan 29, Feb 3 | Numerical integration. Random number generation. |
| **5** | Feb 5, 10 | Ordinary differential equations I. Euler method and advanced difference schemes for ODEs. Nonlinear pendulum. Intro to Chaos. Motion in central field. |
| **6** | Feb 12 | Ordinary differential equations II. Boundary value problems for ODEs. Linear algebra. Quantum particle in a box. |
| **7** | Feb 17, Feb 19 | Basics of fitting and minimization. Chi-square fitting (X-ray spectrum example). |
| **8** | Feb 24, 26 | Bootstrap and Monte Carlo methods. Applications to fitting and image analysis. Mid semester review. |
| **9** | Mar 3, 5 | N-body simulations (star cluster, gas dynamics). Increasing computational efficiency and power. High-performance computing. |
| **Spring break** | Mar 10,12 | **Spring break** |
| **10** | Mar 17, 19 | Time variable signals. Detecting variability. |

| Week | Dates | Topic |
|---|---|---|
| **11** | Mar 24, 26 | Periodicity detection and search for periodic signal. Fourier and its applications to signal processing. |
| **12** | Mar 31, Apr 2 | Unsupervised Machine Learning. Clustering (k-NN algorithm and its relatives) |
| **13** | Apr 7, 9 | Supervised Machine Learning. Information Gain. Entropy. Decision Trees. Examples of use. |
| **14** | Apr 14,16 | Term project discussion. Neural Nets. LLMs. |
| **15** | Apr 21, 23 | Data visualization. |
| **16** | Apr 28, 30 | Term project presentations |

**Attendance**: The students are *expected to attend every class*. If you have to miss >2 classes, you should discuss this with the instructor. Note that if you miss a class, your grade for the in-class quiz/participation will be zero. All requests to have an absence excused must be substantiated with appropriate documentation. Valid excuses include personal illness (with a doctor's note), family emergency, GW sanctioned travel for a team sport, or religious holiday.

**Course Grade:** The final grade will be based on the homework (70%), term project (20%), and attendance, in-class participation, and quizzes (10%). There is no final exam in this course.

## ONLINE RESOURCES

The course webpage will be on Blackboard (http://blackboard.gwu.edu/). Course announcements will be posted on this site. Any distributed reading materials and any PowerPoint slides shown in class will be made available online *after* each class. After entering *Blackboard*, it is necessary for you to click on the course name (PHYS 3181). You are *automatically* subscribed within the *Blackboard* system to the courses for which you are registered (but you also must have a valid GW e-mail address to login).

## Numerical to letter grade conversion

| | | | |
|---|---|---|---|
| ≥ 94.00 | **A** | 70.00 – 73.99 | **C** |
| 90.00 – 93.99 | **A-** | 66.00 – 69.99 | **C-** |
| 86.00 – 89.99 | **B+** | 61.00 – 65.99 | **D+** |
| 82.00 – 85.99 | **B** | 55.00 – 60.99 | **D** |
| 78.00 – 81.99 | **B-** | 50.00 – 54.99 | **D-** |
| 74.00-77.99 | **C+** | ≤ 49.99 | **F** |

# Generative AI Policy for This Course.

**This course is not about learning programming.** There are other courses at GW for that.

**You are allowed and encouraged to Generative AI (GenAI) in ways consistent with the guidelines given below.** However, you are *not* required to use GenAI to succeed in this course.

**Using generative AI implies writing a prompt.** Although you could paste the entire HW problem as a prompt, **do not do this**, at least, not until the very end. Otherwise, you will learn very little and more likely to have an error in the code that will be difficult to track down. Instead, break up the problem into steps and write an algorithm (sequence of steps) which GenAI needs to follow while writing the code for you. Feed this algorithm as a prompt. This helps you to understand the physics of the problem, reduces possibility of mistake in the code written by GenAI, and ensures that it does exactly what you want.

**You are solely responsible for solving the problem.** You cannot say that "AI wrote the code me, I did not know it is wrong or had a mistake". You must fully understand what is happening in the code written by GenAI and be comfortable with what it does and results it produces. Note, you can ask GenAI to explain you pieces of the code it wrote and force it to do additional checks and verifications based on what you expect to see based on your common sense and knowledge of physics. GenAI can help you with commenting the code but it may not do it 100% correctly. *Your milage may vary.*

At the very end, **once you have a working code that solves the problem**, and you are happy with it, you may post the original problem statement as a prompt, see what happens, and compare to the GenAI solution that was produced following the guidelines above. I would appreciate to see your reflections on this comparison as a part of your homework. You will gain +2 extra points for doing so. Note that while I cannot control that you only do it as a last step, there are way of detecting codes produced entirely by GenAI from the same prompt. You may face plagiarism penalty if you are noticed doing this more than once. You will be given a warning first time.

**You can have GenAI do a fancy plotting for you** (including the interactive plots!) if the problem you are solving warrants this. This course does not aim to teach you how to code some advanced plotting and visualizations (although I will show some examples). However, it is good to have nice plots and visualizations and, so, do not hesitate to ask GenAI to help. You just need to be able to explain in words what and how you won't to plot or visualize. The limit is your imagination btu starts with a simple request and then make it more complex.

**Feedback:** During the course I plan to ask you about your experience with GenAI to understand how helpful it is for you. Also, feel free to exchange with your classmates your experiences and best practices in terms of interacting efficiently with GenAI.

**If/when you write code yourself (which I encourage to do too),** you can still ask GenAI to check it for correctness, optimize it (for clarity, structure, execution time, length), and ask GenAI to generate comments for different parts of your code. Of course, ultimately, it is you who are responsible for everything that appears in the code that you turn in.

## University policies

### Academic Integrity Code

Academic integrity is an essential part of the educational process, and all members of the GW community take these matters very seriously. As the instructor of record for this course, my role is to provide clear expectations and uphold them in all assessments. Violations of academic integrity occur when students fail to cite research sources properly, engage in unauthorized collaboration, falsify data, and otherwise violate the Code of Academic Integrity. If you have any questions about whether particular academic practices or resources are permitted, you should ask me for clarification. If you are reported for an academic integrity violation, you should contact Conflict Education and Student Accountability (CESA), formerly known as Student Rights and Responsibilities (SRR), to learn more about your rights and options in the process. Consequences can range from failure of assignment to expulsion from the University and may include a transcript notation. For more information, refer to the CESA website at students.gwu.edu/code-academic-integrity or contact CESA by email cesa@gwu.edu or phone 202-994-6757.

### University policy on observance of religious holidays

Students must notify faculty during the first week of the semester in which they are enrolled in the course, or as early as possible, but no later than three weeks prior to the absence, of their intention to be absent from class on their day(s) of religious observance. If the holiday falls within the first three weeks of class, the student must inform faculty in the first week of the semester. For details and policy, see provost.gwu.edu/policies-procedures-and-guidelines.

### Use of Electronic Course Materials and Class Recordings

Students are encouraged to use electronic course materials, including recorded class sessions, for private personal use in connection with their academic program of study. Electronic course materials and recorded class sessions should not be shared or used for non-course related purposes unless express permission has been granted by the instructor. Students who impermissibly share any electronic course materials are subject to discipline under the Student Code of Conduct. Contact the instructor if you have questions regarding what constitutes permissible or impermissible use of electronic course materials and/or recorded class sessions. Contact Disability Support Services at disabilitysupport.gwu.edu if you have questions or need assistance in accessing electronic course materials.

## Academic support

### Academic Commons

Academic Commons is the central location for academic support resources for GW students. To schedule a peer tutoring session for a variety of courses visit go.gwu.edu/tutoring.
Visit academiccommons.gwu.edu for study skills tips, finding help with research, and connecting with other campus resources. For questions email academiccommons@gwu.edu.

### GW Writing Center

GW Writing Center cultivates confident writers in the University community by facilitating collaborative, critical, and inclusive conversations at all stages of the writing process. Working alongside peer mentors, writers develop strategies to write independently in academic and public settings. Appointments can be booked online at gwu.mywconline.

## Support for students in and outside the classroom

### **Disability Support Services (DSS)** 202-994-8250

Any student who may need an accommodation based on the potential impact of a disability should contact Disability Support Services at disabilitysupport.gwu.edu to establish eligibility and to coordinate reasonable accommodations.

### **Student Health Center** 202-994-5300, 24/7

The Student Health Center (SHC) offers medical, counseling/psychological, and psychiatric services to GW students. More information about the SHC is available at healthcenter.gwu.edu. Students experiencing a medical or mental health emergency on campus should contact GW Emergency Services at 202-994-6111, or off campus at 911.

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## GW Campus Emergency Information

GW Emergency Services: 202-994-6111
For situation-specific instructions, refer to GW's Emergency Procedures guide.

### GW Alert

GW Alert is an emergency notification system that sends alerts to the GW community. GW requests students, faculty, and staff maintain current contact information by logging on to alert.gwu.edu. Alerts are sent via email, text, social media, and other means, including the Guardian app. The Guardian app is a safety app that allows you to communicate quickly with GW Emergency Services, 911, and other resources. Learn more at safety.gwu.edu.

## Protective Actions

GW prescribes four protective actions that can be issued by university officials depending on the type of emergency. All GW community members are expected to follow directions according to the specified protective action. The protective actions are Shelter, Evacuate, Secure, and Lockdown (details below). Learn more at safety.gwu.edu/gw-standard-emergency-statuses.

### Shelter

- Protection from a specific hazard
- The hazard could be a tornado, earthquake, hazardous material spill, or other environmental emergency.
- Specific safety guidance will be shared on a case-by-case basis.

**Action**:

- Follow safety guidance for the hazard.

### Evacuate

- Need to move people from one location to another.
- Students and staff should be prepared to follow specific instructions given by first responders and University officials.

**Action**:

- Evacuate to a designated location.
- Leave belongings behind.
- Follow additional instructions from first responders.

### Secure

- Threat or hazard outside of buildings or around campus.
- Increased security, secured building perimeter, increased situational awareness, and restricted access to entry doors.

**Action**:

- Go inside and stay inside.
- Activities inside may continue.

### Lockdown

- Threat or hazard with the potential to impact individuals inside buildings.
- Room-based protocol that requires locking interior doors, turning off lights, and staying out of sight of corridor window.

**Action**:

- Locks, lights, out of sight
- Consider Run, Hide, Fight

- **Classroom emergency lockdown buttons**
  Some classrooms have been equipped with classroom emergency lockdown buttons. If the button is pushed, GWorld Card access to the room will be disabled, and GW Dispatch will be alerted. The door must be manually closed if it is not closed when the button is pushed. Anyone in the classroom will be able to exit, but no one will be able to get in.

# GW Title IX statement:

The George Washington University (GW) and its faculty are committed to creating a safe and open learning environment for all students. If you or someone you know has experienced sexual harassment, including sexual assault, dating or domestic violence, and stalking, please know that help and support are available. GW strongly encourages all members of the community to take action, seek support, and report incidents of sexual harassment to the Title IX Office. You may contact the Title IX Office at 202-994-7434 or at titleix@gwu.edu or learn more by visiting titleix.gwu.edu. Please be aware that faculty members are required to disclose information about suspected or alleged sexual harassment or other potential violations of the Title IX Sexual Harassment and Related Conduct Policy to the Title IX Office. If the Title IX Office receives information about an incident, they will reach out to offer information about resources, rights, and procedural options as a member of the campus community. Community members are not required to respond to this outreach. If you, or another student you know, wishes to speak to a confidential resource who does not have this reporting responsibility, please contact Counseling and Psychological Services through the Colonial Health Center 24/7 at 202-994-5300, or the Office Of Advocacy and Support at 202-994-0443 or at oas@gwu.edu.